

Isabella DeMarti

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Education

Tecnológico de Costa Rica

B.S. in Industrial Production Engineering. GPA: 88.89/100 | 122 credits completed

Campus Local San Carlos

Expected February 2028

Relevant Coursework: Production Planning & Control, Quality Control, Operations Research, Facility Planning & Design, Supply Chain & Logistics, Probability & Statistics I–II, Industrial Computing, Economic Analysis.

Honors: Academic Honor Scholarship for outstanding academic performance (II-2025, I-2026 & II-2026); International Academy for Quality (IAQ) – Best Quality Control Project Award (2026).

Languages: Spanish (Native); English (Bilingual Proficiency / C1)

Engineering Research Experience

Flexible Manufacturing Cell Project FlexLAB — TEC

San Carlos, Costa Rica

Research and Project Assistant

February 2025 - present

- Identified effective production availability of 16.5 versus 22 planned hours/week by separating setup, breaks, documentation and quality activities, enabling more reliable production planning and OEE measurement.
- Built a Microsoft Power Platform OEE pipeline (Microsoft Lists, Power Apps, Power Query) that replaced roughly three weeks of manual Excel consolidation with automated per-shift reporting.
- Lead-authored three accepted OEE research papers on operational standardization, digitalization, and statistical validation, with presentations in Colombia and future presentations in Panama and Costa Rica.

Additional Experience

Advanced Manufacturing Laboratory SIMTEC — TEC

San Carlos, Costa Rica

Lab Assistant

February 2024 - present

- Operated and supervised advanced manufacturing equipment (CNC lathe, CNC milling machine, laser cutter, 3D scanners, 3D printers) for course student groups.
- Delivered lab introduction and equipment demonstrations to groups of 20+ visiting students from rural areas.
- Authored a CNC lathe operating manual (specifications, control-panel use, safety guidelines, step-by-step procedures), supporting standardized student use and safer equipment operation.

Leadership & Activities

IV International Meeting of Industrial Engineers

San Carlos, Costa Rica

Organizing Committee

August 2025 - April 2026

- Coordinated a two-day international symposium with 21 speakers (6 international) and 100+ attendees.

International Research Presentation

Bogotá, Colombia

Project Representative / Presenter

December 2025

- Lead-authored and presented research on the enablement of OEE measurement through operational standardization, benchmarking Industry 4.0 laboratories to identify improvements for the FlexLAB project.

Projects and Skills

Award-Winning Quality Improvement Project (DMAIC) – Montik Bakery

June 2026

International Academy for Quality (IAQ) Best Quality Control Project Award (2026)

- Applied DMAIC, statistical analysis, FlexSim simulation and economic evaluation to improve a bakery production process.
- Designed a modular dough-portioning mold projected to reduce portioning time by **40%** while maintaining product quality.

Production Planning & Inventory Control System – Almacigos Santa Clara

June 2026

- Developed an Excel-based production planning system incorporating stage-specific yield-loss modeling, KPI tracking and predictive order-risk alerts.

Facility Layout & Capacity Expansion – Almacigos Santa Clara

June 2026

- Applied Systematic Layout Planning (SLP) to redesign production flow, increasing projected plant capacity by **20%** without capital investment.

Additional projects: Warehouse Management System (Excel/VBA), Financial Evaluation of FIFCO (FNE, VPN, TMAR), Queueing Theory Analysis (POM-QM).

Technology: Power BI, Power Query (M), Power Apps (Power Fx), Microsoft Lists, Minitab, Excel/VBA, SolidWorks.

Certifications: Certified SOLIDWORKS Associate (CSWA), Dassault Systèmes; Lean Six Sigma Yellow Belt (MoreSteam, in progress)